The University of Hertfordshire
School of Physics, Engineering and Computer Science

Coursework: Final Coursework

MSC Computer Science (online)
7WCM2034-0911-2026 - Fundamentals of Software Engineering

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I confirm that this submission is my own work and conforms to the university's academic integrity policies.

1.

Software engineering is important because we rely on software for things like banking, healthcare and communication. A mistake in one of these systems could delay a service or put important information at risk. It helps teams plan, build and test software so it works properly and meets people's needs. The team can agree on what the software should do before they write the code. It also makes software easier to fix and improve later. It helps people work on the same project without losing track of changes.

2.

a. Systems Architect
b. Service Manager
c. Quality Assurance Manager
d. ICT Trainer
e. ICT Security Specialist
f. ICT Security Manager
g. ICT Operations Manager
h. Test Specialist

(CEN, 2012)

3.

a) Software requirements describe the needs and limits a software product must meet to help solve a real problem. They should also be possible to check, so we can tell whether the software meets them (Bourque and Fairley, 2014).

b) Functional requirements say what the software must do, such as formatting text. Nonfunctional requirements set limits on the software, such as how fast, reliable or secure it must be (Bourque and Fairley, 2014).

c) SWEBOK lists seven software design principles (Bourque and Fairley, 2014):

1. Abstraction means focusing on the details that matter and leaving out those that do not.
2. Coupling and cohesion describe how much parts depend on each other and how closely the work within each part belongs together.
3. Decomposition and modularization mean dividing a large system into smaller parts with clear connections.

4. Encapsulation and information hiding mean keeping a part's internal details hidden from other parts.
5. Separation of interface and implementation means showing what a part does without exposing how it works inside.
6. Sufficiency, completeness and primitiveness mean including what a part needs, leaving out what it does not, and using simple building blocks.
7. Separation of concerns means looking at different design issues separately so they are easier to manage.

4.

The software development life cycle has five main stages. Each stage has related areas of software engineering and people who do the work. Some roles work across more than one stage.

1. Requirements gathering and analysis. The main area is requirements engineering. Business Analysts identify business needs, while Systems Analysts turn them into software requirements.
2. Design. The main area is software design. Systems Architects plan the system, while Developers and Database Administrators help design its parts and data.
3. Implementation. The main areas are software construction and software configuration management. Developers write the code, while Project Managers organise the work.
4. Testing. The main areas are software testing and software quality. Test Specialists test the system, while Quality Assurance Managers check that quality rules are followed.
5. Deployment and maintenance. The main areas are software maintenance and software configuration management. Systems Administrators install and update systems. Service Managers, Technical Specialists and Service Desk Agents help keep them working and support users.

(CEN, 2012).

5.

My chosen profession is Service Desk Agent as per my student ID last digit, 4. This is usually the first person users contact when they have a technical problem, often by phone or email. The agent listens, records the issue, tries to find the cause and explains what the user can do. If the problem cannot be solved at the service desk, the agent passes it to a specialist and follows it until it is resolved.

This role is most active after the software has been deployed. The problems users report can also help the team plan changes for a future version. Design, coding and testing are mainly done by other specialists. The Service Desk Agent helps people use the software and supports it while it is in use (CEN, 2012).

6.

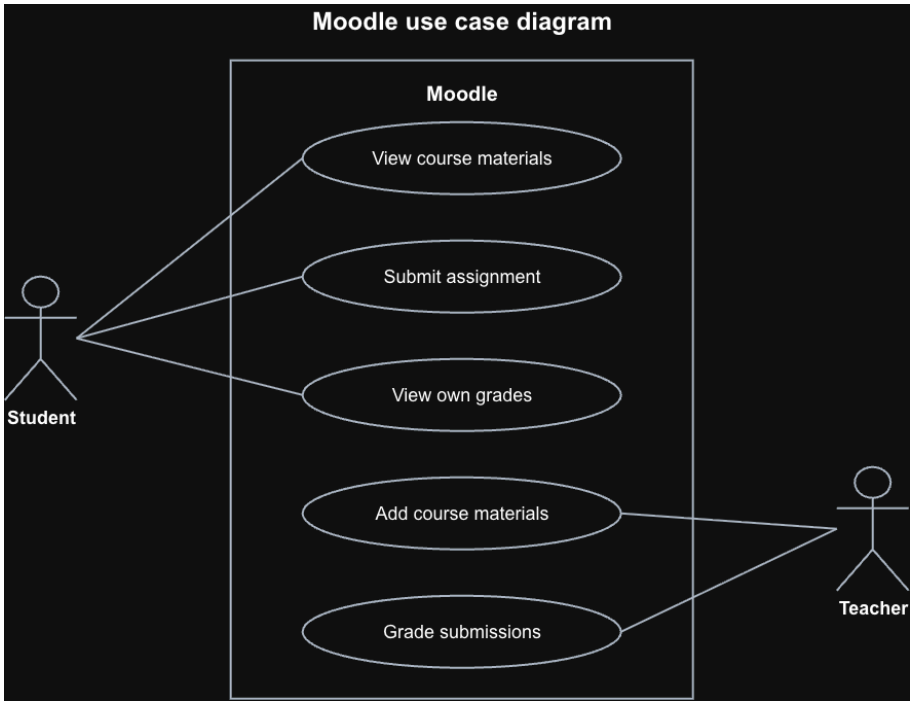
Agile principle 4, as per the last digit of my student ID:

a) Business people and developers should talk and work together every day. This helps developers understand what the business needs and lets business people see how the work is going.

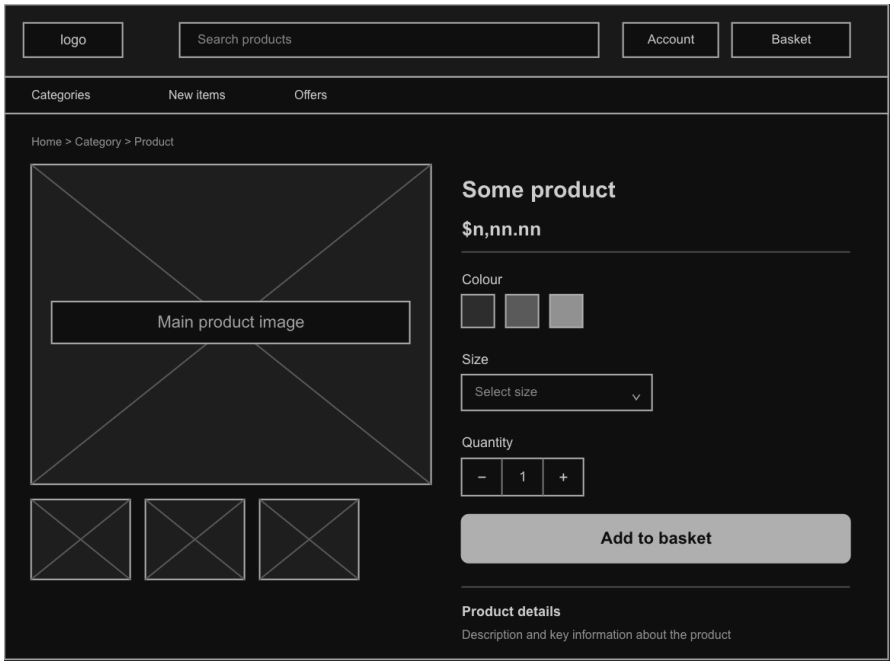
b) Regular contact helps the team clear up questions early. It also makes it easier to change the software when the business needs something different.

c) No other principle says exactly the same thing. Principle 6 is related because it also focuses on communication, but it is about how information is shared within a development team (Beck et al., 2001).

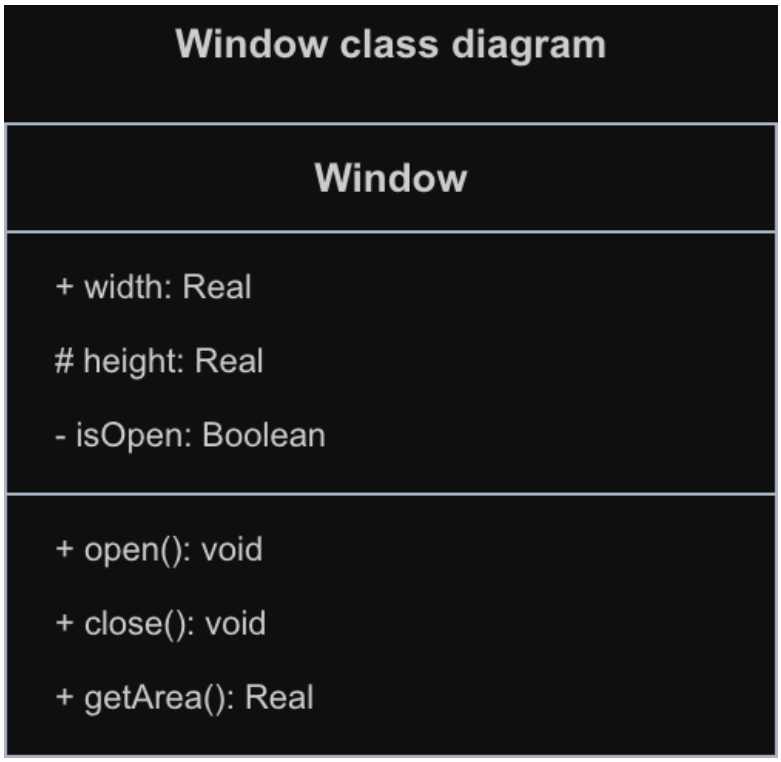
7.



8.


logo
Search products
Account
Basket
Categories
New items
Offers
Home > Category > Product
Main product image
Some product
$n,nn.nn
Colour
Size
Select size
Quantity
−
1
+
Add to basket
Product details
Description and key information about the product

9.


Window class diagram
Window
+ width: Real
height: Real
- isOpen: Boolean
+ open(): void
+ close(): void
+ getArea(): Real

10.

a) CodeIgniter mainly uses the Model View Controller (MVC) pattern. The model handles data, the view shows the page, and the controller handles requests and connects them.
b) This gives each part a clear job. It makes code easier to find, change and maintain because the data and page display are kept separate (CodeIgniter, no date).

11.

```
/// This class does some basic maths and text work.

public class MyClass {

    /// This adds the numbers a and b.

    /// It gives back their total.

    public int add(int a, int b) {

        return a + b;

    }


    /// Here a and b are two pieces of text.

    /// It joins them and gives back one piece of text.

    public String add(String a, String b) {

        return a + b;

    }


    /// This takes b away from a.

    /// It gives back the answer.
```

```
public int sub(int a, int b) {
    return a - b;
}

/// This multiplies a by b.
/// It gives back the answer.
public int mult(int a, int b) {
    return a * b;
}

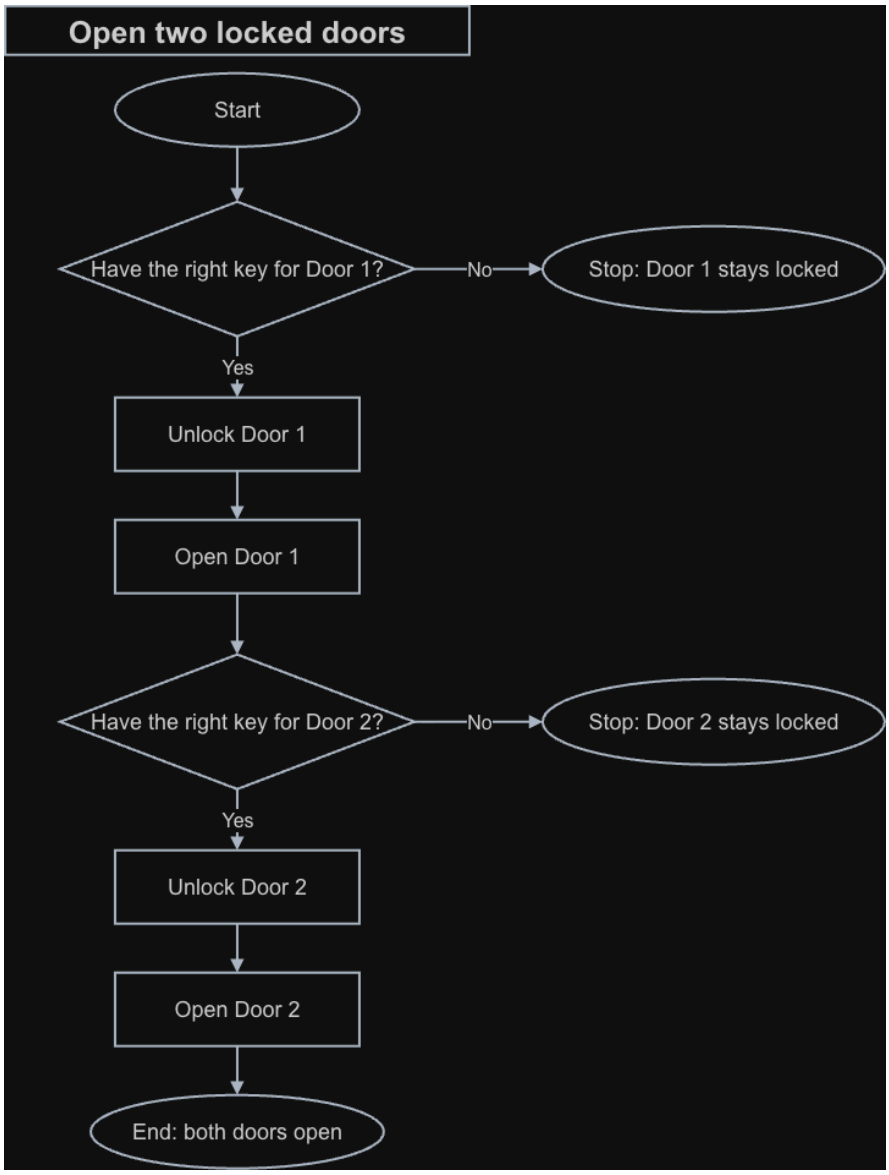
/// This divides a by b and gives back a whole number.
/// b cannot be zero.
public int div(int a, int b) {
    return a / b;
}

/// This divides a by b and gives back the remainder.
/// b cannot be zero.
public int mod(int a, int b) {
    return a % b;
}

/// This puts Hello straight before the name given.
```

```
    /// It gives back the greeting as text.
    public String sayHello(String name) {
        return "Hello" + name;
    }
}
```

12.



13.

https://github.com/ravshan2rtikov/uhf-se-coursework.git